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MILITARY SPECIFICATION

# VALVE, SOLENOID OPERATED

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification establishes the requirements for a direct acting, poppet style, two-way shutoff solenoid operated valve for use on a military ground vehicle.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS
MILITARY

| | |
|---|---|
| MIL-H-5606 | - Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance. |
| MIL-H-6083 | - Hydraulic Fluid, Petroleum Base for Preservation and Operation. |

STANDARDS
MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-202 | - Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-454 | - Standard General Requirements for Electronic Equipment. |
| MIL-STD-810 | - Environment Test Methods. |
| MIL-STD-1246 | - Product Cleanliness Levels and Contamination Control Program. |
| MIL-STD-1275 | - Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles. |
| MIL-STD-1523 | - Age Control of Age-Sensitive Elastomeric Material. |
| MIL-STD-45662 | - Calibration Systems Requirements. |

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

| | |
|---|---|
| 12265543 | - Valve, Solenoid Operated. |

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish solenoid valves which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent valves delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.



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3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Materials shall be free from defects which adversely affect performance or serviceability of the finished product (see 4.6.1 and 6.3).

3.2.1 Material compatibility. Materials for the solenoid valve shall be compatible with hydraulic fluid per MIL-H-5606 or MIL-H-6083.

3.2.2 Age-sensitive materials. The use of age-sensitive materials shall be controlled by MIL-STD-1523. Any age-sensitive material shall be noted on the manufacturer's working drawing.

3.2.3 Metals. Metals shall be corrosion resistant or suitably treated to prevent corrosion. The use of dissimilar metals, especially brass, copper, or steel in intimate metal to metal contact, shall be avoided.

3.3 Design and construction. The solenoid valve shall be a direct acting, poppet style shutoff valve, normally closed. The solenoid must be energized for the valve to remain open and allow flow of fluid. The solenoid valve shall conform to Drawing 12265543 and other applicable drawings, specifications, and standards (see 4.6.1 and 4.6.2).

3.3.1 Operating fluid. The valve shall operate satisfactorily with hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083.

3.4 Performance.

3.4.1 Pressure rating (internal).

3.4.1.1 Proof pressure. The proof pressure of the valve shall be 4500 (+ 100, -0) pounds per square inch (psi) (see 4.6.3.1).

3.4.1.2 Burst pressure. The burst pressure of the valve shall be 7500 (+ 100, -0) psi (see 4.6.3.2).

3.4.2 Solenoid rating.

3.4.2.1 Pull-in voltage. The valve shall actuate at any voltage applied to the solenoid within a range between 18 and 30 volts direct current (V dc) (see 4.6.4.1).

3.4.2.2 Hold-in voltage. Once the valve is actuated, the valve shall not release at any voltage to the solenoid greater than 16 V dc (see 4.6.4.2).

3.4.2.3 Coil current. Actuating current shall not exceed 0.51 ampere (A) dc at 30 V dc and 77 degrees Fahrenheit (°F) [25 degrees Celsius (°C)] (see 4.6.4.3).

3.4.2.4 Insulation resistance. With 500 V dc applied between the coil terminals connected together and the case, there shall be at least 100 megohms of resistance (see 4.6.4.4).



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3.4.2.5 Dielectric withstanding voltage. The solenoid valve shall be capable of initially withstanding 500 volts root mean square (rms), 60 Hertz (Hz) for 1 minute, or 600 V rms, 60 Hz for 1 second between terminals tied together and the case (see 4.6.4.5).

3.4.2.6 Voltage transient. The valve shall perform as specified when subjected to transient voltages as specified in MIL-STD-1275 (see 4.6.4.6).

3.4.3 Rated flow. The flow capacity of the valve shall be not less than 0.25 gallon per minute (gpm), inlet to outlet, with a pressure drop across the valve of 75 psi (see 4.6.5).

3.4.4 Internal leakage. With the solenoid de-energized and the valve closed, the leakage, inlet to outlet, shall not exceed 0.25 milliliter (ml) after 2 minutes at 3000 psi (see 4.6.6).

3.4.5 External leakage. There shall be no evidence of leakage through external seals (see 4.6.7).

3.4.6 Response time.

3.4.6.1 Time to open. The time to establish flow, after the solenoid shown in the drawing test circuit has been energized, shall not be more than 1 second (see 4.6.8.1).

3.4.6.2 Time to close. The time to terminate flow, after the solenoid shown in the drawing test circuit has been de-energized, shall not be more than 1 second (see 4.6.8.2).

3.4.7 Environmental conditions. The design, materials, and construction of the valve shall enable the unit to operate in the environmental conditions called out herein. The valve shall meet the performance requirements under the following temperature, shock, and vibration conditions.

3.4.7.1 Body and ambient temperature.

3.4.7.1.1 High temperature. The solenoid valve shall operate when subjected to a high ambient temperature of 140°F (60°C). The valve unit shall not be damaged nor its performance impaired after storage in a non-operating mode at a high ambient temperature of 158°F (70°C) (see 4.6.9.1.1).

3.4.7.1.2 Low temperature. The solenoid valve shall operate when subjected to a low ambient temperature of minus 26°F (minus 32°C). The valve unit shall not be damaged nor its performance impaired after storage in a non-operating mode at a low ambient temperature of minus 65°F (minus 54°C) (see 4.6.9.1.2).

3.4.7.2 Hydraulic fluid temperature. The solenoid valve shall operate using hydraulic oil conforming to MIL-H-5606 or MIL-H-6083 at operating oil temperatures of minus 26°F (minus 32°C) to 170°F (77°C) with ambient air conditions as specified herein (see 4.6.9.2).



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3.4.7.3 Shock (non-operating). The units shall operate after exposure to a shock impulse of 25 gravity units (g) for 11 milliseconds (ms), half-sine wave form (see 4.6.9.3).

3.4.7.4 Vibration (non-operating). The solenoid valve shall operate as specified herein during and after exposure to vibration conditions (see 4.6.9.4):

a. Simple harmonic motion.
b. Motion applied along three mutually perpendicular directions.
c. Peak acceleration of ± 10 g at all frequencies.
d. Frequency range 20 to 2000 Hz.
e. Frequency sweep varied logarithmically minimum to maximum only.
f. Sweep time 35 minutes ± 5 minutes.

3.4.7.5 Salt-fog (non-operating). The solenoid valve shall operate after being exposed to a salt laden environment (see 4.6.9.5).

3.4.7.6 Fungus. The solenoid valve shall be constructed of only materials that are fungus non-nutrients as identified in MIL-STD-454, requirement 4 where possible. Fungus growth shall not degrade performance (see 4.6.9.6).

3.5 Cleanliness. Upon completion of testing, a sufficient quantity of fluid shall be retained in the valve to assure proper preservation during extended storage and shipment. Ports shall be covered with suitable plugs. The internal cleanliness of the valve shall conform to level 300 of MIL-STD-1246, except measurement of cleanliness level paragraph 5.1.2 need not be conducted providing the following conditions are met (see 4.6.2 and 4.6.10):

a. All parts are cleaned or maintained clean including the final assembly in accordance with MIL-STD-1246 provisions.

b. During any application of fluid, the final assembly is filtered in accordance with paragraph 4.6 of this specification.

3.6 Marking. Marking of the solenoid valves shall conform to MIL-STD-130 and drawing 12265543 (see 4.6.11).

3.7 Workmanship. Workmanship shall be of such quality as to assure that solenoid valves furnished under this specification are free from defects that compromise, limit, or reduce performance in intended use (see 4.6.12).



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions:

| | | |
|---|---|---|
| a. | Air temperature | 73.4°F ± 32°F (23°C ± 10°C) |
| b. | Barometric pressure | 725 + 50mm Hg<br>- 75 |
| c. | Relative humidity | 50 ± 30 percent |

4.4 First article inspection. First article inspection shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply solenoid valves that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of 2 solenoid valves. Preproduction inspection shall consist of inspection as specified in table I.



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TABLE I. Classification of inspection.

| Title | Requirement | Inspection | First article | Quality conformance | |
|---|---|---|---|---|---|
| | | | | Examination | Tests |
| Materials and construction | 3.2.3.3 | 4.6.1 | X | | X |
| Defects | 3.3.3.5 thru 3.7 | 4.6.2 | X | X | |
| Proof pressure | 3.4.1.1 | 4.6.3.1 | X | | X |
| Burst pressure | 3.4.1.2 | 4.6.3.2 | X | | |
| Pull-in voltage | 3.4.2.1 | 4.6.4.1 | X | | X |
| Hold-in voltage | 3.4.2.2 | 4.6.4.2 | X | | X |
| Coil current | 3.4.2.3 | 4.6.4.3 | X | | X |
| Insulation resistance | 3.4.2.4 | 4.6.4.4 | X | | X |
| Dielectric withstanding voltage | 3.4.2.5 | 4.6.4.5 | X | | X |
| Voltage transient | 3.4.2.6 | 4.6.4.6 | X | | X |
| Rated flow | 3.4.3 | 4.6.5 | X | | X |
| Internal leakage | 3.4.4 | 4.6.6 | X | | X |
| External leakage | 3.4.5 | 4.6.7 | X | | X |
| Time to open | 3.4.6.1 | 4.6.8.1 | X | | X |
| Time to close | 3.4.6.2 | 4.6.8.2 | X | | X |
| High temperature | 3.4.7.1.1 | 4.6.9.1.1 | X | | X |
| Low temperature | 3.4.7.1.2 | 4.6.9.1.2 | X | | X |
| Hydraulic fluid temperature | 3.4.7.2 | 4.6.9.2 | X | | X |
| Shock | 3.4.7.3 | 4.6.9.3 | X | | |
| Vibration | 3.4.7.4 | 4.6.9.4 | X | | |
| Salt-fog | 3.4.7.5 | 4.6.9.5 | X | | |
| Fungus | 3.4.7.6 | 4.6.9.6 | X | | |

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select 2 solenoid valves, from the first 10 valves produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table I.

4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.



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4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the solenoid valves of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective.

| Classification | AQL |
|---|---|
| Major | 1.0 |
| Minor | 2.5 |

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

| Category | Defect | Method of inspection |
|---|---|---|
| Critical | None | |
| Major | AQL 1.0% Defective | |
| 101 | Dimensions, out of tolerance (see 3.3) | SIE 1/ |
| 102 | Cleanliness, inadequate (see 4.6.10). | Visual |
| Minor | AQL 2.5% Defective | |
| 201 | Identification marking, improper (see 4.6.11). | Visual |
| 202 | Workmanship, faulty (see 4.6.12). | Visual |

1/ SIE= Standard Inspection Equipment.

4.5.3 Tests. All solenoid valves shall be subjected to quality conformance tests specified in table I.



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4.5.4 Failure. Failure of any valve to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection. Unless otherwise specified, inspections shall be conducted under the following conditions:

| | |
|---|---|
| Hydraulic fluid | MIL-H-5606 or MIL-H-6083 |
| Fluid temperature | 73.4 ± 32°F (23 ± 10°C) |
| Fluid Supply filliation | 10 micron (25 micron absolute) |
| Test circuit | Per drawing 12265543 |

4.6.1 Materials and construction. Conformance to 3.2 and 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3 and 3.5 through 3.7 shall be determined by examination for the defects listed in Table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Pressure rating. Pressure tests shall be performed with the valve in the closed configuration and the outer port plugged.

4.6.3.1 Proof pressure. To determine conformance to the requirements of 3.4.1.1, a proof pressure of 4,500 psi shall be applied to the inlet port at a rate of 25,000 psi per minute and held for a period of 2 minutes. There shall be no permanent deformation, rupture, or measurable external leakage.

4.6.3.2 Burst pressure. To determine conformance to the requirements of 3.4.1.2, a burst pressure of 7,500 psi shall be applied to the inlet port at a rate of 25,000 psi per minute and held for a period of 2 minutes. There shall be no permanent deformation, rupture, or measurable external leakage. Solenoid valves subjected to this test shall be indelibly marked as damaged and shall not be delivered as acceptable units.

4.6.4 Solenoid rating.

4.6.4.1 Pull-in voltage. To determine conformance to 3.4.2.1, the solenoid shall be operated to determine the minimum pull-in voltage with the inlet at maximum operating pressure and with the discharge restricted to maintain rated flow. To verify pull-in repeat the test using 30 V dc.



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4.6.4.2 Hold-in voltage. To determine conformance to 3.4.2.2, the valve shall be tested by energizing it with greater than 24 V dc at maximum operating pressure and the discharge restricted to maintain flow. The hold-in voltage shall be reduced to 16 V dc and the valve shall not be released.

4.6.4.3 Coil current. To determine conformance to 3.4.2.3, the solenoid shall be activated and the actuating current shall be measured.

4.6.4.4 Insulation resistance. To determine conformance to the requirements of 3.4.2.4, the units shall be tested in accordance with MIL-STD-202, method 302, condition B.

4.6.4.5 Dielectric withstanding voltage. To determine conformance to the requirements of 3.4.2.5, the units shall be tested in accordance with MIL-STD-202, method 301. The test shall be conducted at 400 V rms, 60 Hz.

4.6.4.6 Voltage transient. To determine conformance to the requirements of 3.4.2.6, the units shall be tested in accordance with MIL-STD-1275.

4.6.5 Rated flow. To determine conformance to the requirements of 3.4.3, the units shall be tested with the solenoid energized to establish a flow of not less than 0.25 gpm at 75 psi inlet to outlet.

4.6.6 Internal leakage. To determine conformance to the requirements of 3.4.4, the unit shall be tested with the solenoid de-energized and valve closed. A pressure of 3,000 psi shall be applied to the inlet port and held for a period of 2 minutes. Internal leakage at the outlet shall be verified not to exceed 0.25 ml after 2 minutes.

4.6.7 External leakage. To determine conformance to the requirements specified in 3.4.5, visual examination of the units shall verify no external leakage during conformance verification.

4.6.8 Response time.

4.6.8.1 Opening time. To determine conformance to the requirements specified in 3.4.6.1, the units shall open in not more than 1 second by establishing flow at a rate of 0.25 gpm at 3000 psi after the solenoid shown in the drawing test circuit has been energized.

4.6.8.2 Closing time. To determine conformance to the requirements specified in 3.4.6.2, the units shall close in not more than 1 second by terminating flow after the solenoid shown in the drawing test circuit has been de-energized.



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4.6.9 Environmental conditions.

4.6.9.1 Body and ambient temperature.

4.6.9.1.1 High temperature. To determine conformance to 3.4.7.1.1, the solenoid valve shall be tested in accordance with MIL-STD-810, method 501.2, procedures I and II except as follows:

| | |
|---|---|
| Operating temperature: | 140°F (60°C). |
| Non-operating temperature: | 158°F (70°C). |

4.6.9.1.2 Low temperature. To determine conformance to 3.4.7.1.2, the solenoid valve shall be tested in accordance with MIL-STD-810, method 502.2, procedures 1 and II except as follows:

| | |
|---|---|
| Operating temperature: | minus 26°F (minus 32°C). |
| Non-operating temperature: | minus 65°F (minus 54°C). |

4.6.9.2 Hydraulic fluid temperature. To determine conformance to 3.4.7.2, the solenoid valve shall be tested in accordance with MIL-STD-810, method 501.2, procedure II and method 502.2 procedure II. This test may be run in conjunction with 4.6.9.1.1 and 4.6.9.1.2 tests.

4.6.9.3 Shock (non-operating). To determine conformance to the requirements of 3.4.7.3, the units shall be subjected to condition J, method 213 B, MIL-STD-202. A peak acceleration of 25g shall apply.

4.6.9.4 Vibration (non-operating). To determine conformance to the requirements of 3.4.7.4, the solenoid valve shall be subjected to condition C, method 204C, MIL-STD-202. In addition to the frequency sweep, vibrate 5 minutes at each of two observed resonant frequencies.

4.6.9.5 Salt-fog (non-operating). To determine conformance to the requirements of 3.4.7.5, the unit shall be tested in accordance with procedure I, method 509.2, MIL-STD-810.

4.6.9.6 Fungus. To determine conformance to 3.4.7.6, a certification of compliance with supporting data shall be presented by the manufacturer, or a valve shall be tested in accordance with MIL-STD-810 method 508.3.

4.6.10 Cleanliness. To determine conformance to the requirements of 3.5, the valves shall be examined in accordance with MIL-STD-1246 and table II of this specification.

4.6.11 Marking. To determine conformance to 3.6, the solenoid valve shall be visually examined for conformance to MIL-STD-130 amd drawing 12265543.

4.6.12 Workmanship. Workmanship shall be performed at all phases of fabrication and testing to verify conformance to 3.7.



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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with the applicable packaging standard or packaging data sheet for the desired level of protection specified (see 6.2).

6. NOTES

6.1 Intended use. The solenoid valve furnished under this specification is to be used in a military ground vehicle electron drive system as an emergency bypass valve.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article samples are not required (see 3.1).
c. If responsibility for inspection shall be other than as specified (see 4.1).
d. If responsibility for test equipment shall be other than as specified (see 4.1.1).
e. If inspection conditions shall be other than as specified (see 4.3).
f. Preproduction inspection, if required (see 4.4.1).
g. Initial production inspection, if not required (see 4.4.2).
h. Selection of applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of applicable material specifications without jeopardizing intended use of the item shall be encouraged (see 3.2).

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